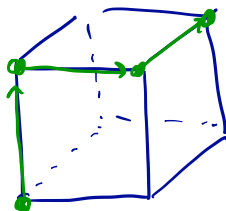


# The Simplex Algorithm

The picture:



## Adjacent Edges and Bases — 1/2

$x \in \text{Ext } \underline{P}$  :  $B$  basis,  $B \subseteq [n]$ ,  $|B|=m$

$$x = \begin{pmatrix} x_B \\ 0 \end{pmatrix}, \quad x_B = A_B^{-1} b, \quad A_B = m \text{ columns of } A$$

corresponding to matrix of  $B$

$$\text{value: } c^T x = c_B^T x_B + c_{B^c}^T 0 = c_B^T x_B = c_B^T A_B^{-1} b.$$

Move to adjacent Ext pt : pick  $\hat{i} \notin B$ .

$$\underline{B_{\text{new}} \subseteq B \cup \{\hat{i}\}}$$

Must determine: which elt of  $B$  leaves?

## Adjacent Edges and Bases — 2/2

$$x_{\text{new}} = \underline{x + \theta d}$$

$$d_i = 1, \quad d_j = 0 \quad \forall j \in (B \cup \hat{i})$$



$$\Rightarrow d = \begin{pmatrix} d_B \\ 1 \\ 0 \end{pmatrix} \leftarrow \hat{i} \text{ location.}$$

$$Ax_{\text{new}} = b: \quad \cancel{Ax} + \theta \cancel{Ad} = \cancel{b}$$

$$\Rightarrow \underline{Ad = 0}: \quad A_i \hat{i} + A_B d_B = 0$$

$$\Rightarrow \boxed{d_B = -\bar{A}_B^{-1} A_i \hat{i}}$$

$$x_{\text{new}} = \begin{pmatrix} x_B \\ 0 \\ 0 \end{pmatrix} + \theta \begin{pmatrix} -\bar{A}_B^{-1} A_i \hat{i} \\ 1 \\ 0 \end{pmatrix}$$

What is  $\theta$ ?

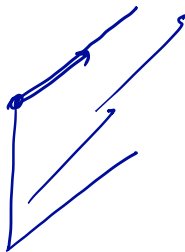
$\geq 0$

## How Far to Move — 1/2

$$\theta \geq 0 \quad \left[ x_B - \theta \overbrace{A_B^{-1} A_i}^{d_B} \right] \geq 0$$

$$Ax_{\text{new}} = b.$$

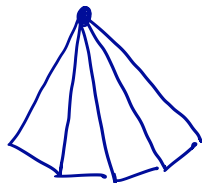
$$\theta = \min_{\substack{j \in B \\ d_j > 0}} \frac{x_j}{d_j}$$



①  $\theta > 0$  if finite :  $x_j > 0, d_j > 0$

②  $+\infty$  if all  $d_j \leq 0$ .

③ 0 if  $x_j = 0, d_j > 0$ .



# How Far to Move 2/2

$$a_i x \leq b_i$$

$$a_i x \leq b_i \quad i=2 \dots 6$$

$$a_i x + s_i = b_i \quad | \quad 6$$

$$a_i x + s_i = b_i \quad | \quad =$$

$$x, s \geq 0$$

9 variables

$$s_2 = \dots = s_6 = 0$$

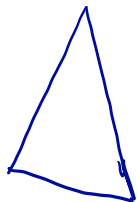
$$B \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ s_1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$|B| = 6$$

" $x_B$ "

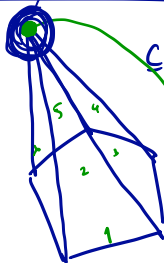
$x$

$\mathbb{R}^3$



$$Ax = b$$

$$A \text{ } m \times n \quad |b| = m$$



6 total ineq.

5 tight eq. const.

2, 3, 4, 5, 6

$$Ax = b$$

$$x \geq 0$$

$$A \quad m \times n, \quad m \leq n$$

The diagram shows a matrix  $A$  represented as a rectangle. A vertical line is drawn through the first column of  $A$ , and the label  $A_i$  is placed above this line. To the right of  $A$ , there is a vertical vector labeled  $x_B$  at the top, followed by a horizontal line, and then the vector  $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ . This is followed by an equals sign and a column vector  $\begin{bmatrix} b \end{bmatrix}$ .

$$Ax = \sum A_i x_i$$

$$A = \begin{bmatrix} A_B & \vdots & A_{B^c} \end{bmatrix} \begin{bmatrix} x_B \\ \hline 0 \end{bmatrix} = A_B x_B + A_{B^c} 0 = A_B x_B.$$

Post-class:

A specific example where  $\theta = 0$ .

If  $B = \{5, 6, 7\}$ ,

$$A_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x_1 - 2x_2 - x_3 + 3x_4 + x_5 + 0x_6 + 0x_7 = 0$$

$$\frac{1}{2}x_1 - 10x_2 - 2x_3 + 4x_4 + 0x_5 + x_6 + 0x_7 = 0$$

$$0x_1 + 0x_2 + x_3 + 0x_4 + 0x_5 + 0x_6 + x_7 = 1$$

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0.$$

$m=3, n=7$ . For an extreme point we need  $n-m=4$  of the  $x_i$ 's to be equal to zero.  $|B|=m=3$ .

Let  $B = \{5, 6, 7\}$ . Then  $x_1 = x_2 = x_3 = x_4 = 0$ , and  $x_B = \begin{pmatrix} x_5 \\ x_6 \\ x_7 \end{pmatrix} = A_B^{-1} b = I \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$\Rightarrow x_B = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}_B$ . Say we want to introduce  $\hat{z}=1$

into the basis. Then:  $x_{\text{new}} = x + \theta d$ , where

$$d = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad d_B = -A_B^{-1} \begin{pmatrix} 1 \\ \frac{1}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -\frac{1}{2} \\ 0 \end{pmatrix} \Rightarrow x_{\text{new}} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \theta \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\frac{1}{2} \\ 1 \end{pmatrix}$$

- continued -

We have:

$$x_{\text{new}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \theta \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ -1/2 \\ 0 \end{pmatrix}$$

We need  $x_{\text{new}} \geq 0$

$$\Rightarrow \theta = 0 \quad \text{b/c of } 5^{\text{th}} \text{ \& } 6^{\text{th}} \\ \text{coordinates.}$$

When multiple bases describe the same extreme points it  
 $\Rightarrow$  called DEGENERACY.